

No. of Functional Features Included in the Solution

Date	02 July 2026
Team ID	ghousiyaunnissa@gmail.com
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	CSV Data Ingestion Engine	Automated structural reading and feature array loading of transaction histories.	creditcard.csv Data Layer	Done	0.5
2	Feature Matrix Scaling	Normalization of numerical financial records to isolate evaluation outliers.	scaler.pkl Preprocessing	Done	0.5
3	Random Forest Ensemble	Parallelized multi-tree ensemble framework matching mathematical prediction weights.	train.py Model Logic	Done	1.0
4	Binary Weight Export	Secure file serialization of model architecture coordinates via pickle interface.	fraud_model.pkl Serialization	Done	0.5
5	Flask Server Core	Setting up background web engine instance parameters and request handlers.	app.py Backend Server	Done	1.0
6	Multi-Field Entry Form	Clean user interface input grid designed to capture user parameters.	templates/ UI View	Done	0.5
7	Real-Time API Prediction	Instant computational pipeline calculation matching input data weights.	app.py Core Controller	Done	0.5
8	Dynamic Status Indicators	Visual validation panels transforming colors according to prediction risk.	static/ Interface Style	Done	0.5

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Multi-Field Entry Form, Dynamic Status Color Indicators
2	Backend / Logic	3	Flask Server Routing Engine, Random Forest Voting Pipeline, Binary Weights Export
3	Database / Storage	1	Local File Ingestion Handling Engine (creditcard.csv layer)
4	API / Integration	1	Scikit-Learn Preprocessing Object Matrix Linking Pipeline
5	Security / Authentication	1	Secure Runtime Input Validation (Null/NaN matrix error block protection)